

controller to monitor energy usage, basically kWh consumed, kWh stored and kWh generated/exported. It can also communicate with industry-standard energy meters for billing and tariff setting purposes.

Major power management blocks. The DC-DC converter block is fed from DC bus. Depending on type of vehicle connected and demand raised by the vehicle battery management system for required voltage and current, the central controller configures the DC-DC converter over the communication bus. This option is typically for DC fast charging, and multiple DC-DC converter modules would work in tandem to fulfill the load.

The DC-AC inverter is also fed from DC bus, but this caters to vehicles that can only accept AC to charge or for general slow-charging applications. The bi-directional inverter serves two purposes. It either feeds DC bus to fulfill demand or it exports power back to the grid, when the charging station is lying idle or is needed to supplement the grid during peak hours. Key figures of merit for any power conversion block these days are:

- Very high efficiency: > 95 per cent end-to-end are now realistic figures
- Very high power densities: smaller and even smaller systems, as real estate is a significant deployment cost

Both of the above points are met using advancements in silicon. Wide-band-gap semiconductors, especially silicon-carbide devices, are able to work at very high switch-

ing frequencies, at much higher junction temperatures and with higher efficiencies. In addition to this, there is an automatic reduction in size of passive components like magnets and capacitors. Better magnetic materials also lend to smaller and low-loss designs, handling much higher powers.

The master central controller is the brain of the station. It performs functions starting from identifying and engaging the user/subscriber till ensuring the vehicle is charged in an optimal manner. It is a powerful combination of high-performance computing, connectivity and sensing. Major functionalities are:

- **User ID and payment.** This is the most visible functionality as far as the user is concerned. This is done through a smart card, OTP, NFC-enabled phones or even Bluetooth. All these sub-systems are controlled by the MPU/MCU on board.

- **Power management.** This is the most critical yet invisible part of the station. The system controller continually monitors the power scenario—supply and demand. Then it decides how to fulfill demand, from supply—whether solar alone is able to supply the load, or a combination of solar and storage is needed, or it needs partial input from the grid as well. There may be scenarios where there could be excess availability or excess demand. It is intelligent enough to route power correspondingly, by altering settings of the various power blocks described earlier.

- **Connectivity.** These days, stations and deployments need to be connected to the cloud for remote

monitoring and control. These have to talk to the central management system (CMS) periodically, to report transactions, parametric, diagnostics and operational data. These also need to take operational commands and settings from the CMS. So, a multitude of connectivity options, both wired and wireless, is provided. 3G/4G, Wi-Fi, Ethernet and even LoRa have been used for remote monitoring.

- **Protections, diagnostics and fault reporting.** The system to prevent malfunction has fast-acting protection mechanisms that may be triggered by external events like surges or lightning strikes, or due to operational issues, accidental or deliberate misuse/abuse or from short-circuits, over-temperature or over-voltage/over-current conditions. To keep operational costs low and to have minimum downtime, systems can self-report issues that may arise from time to time. Modular build allows pinpointing which faulty section needs to be replaced at the field, so the technician can arrive well prepared.

A brief explanation of how a solar EV charging system may be deployed is given in this article. It is also possible to customise designs as per needs of the OEM. e-Mobility and EV charging infrastructure are some of the key focus areas, and relentless research is being done to address high performance in all functional blocks. End-to-end silicon for realising realistic EV charging stations is available, along with many reference designs, to keep the time to market short. **EFY**

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